23秋 线性代数期末试题（回忆版）

2. Fill the in blanks.

(1) Let u be a n-dimensional nonzero real column vector and $A=I+uu^T$, where I is the nxn identity matrix. Then all the distinct eigenvalues of A are _____ with algebraic multiplicties _____.

(2) The singular values of the matrix $A=\begin{bmatrix}1&-2\\-1&1\\2&1\end{bmatrix}$ are _____.

(3) Let $A=\begin{bmatrix}2&-1&0&0&0\\4&-2&0&0&0\\0&0&1&1&0\\0&0&0&1&1\\0&0&0&0&1\end{bmatrix}$. Then $A^{2024}=$ _____.

(4) Let x_n be a sequence defined by $x_0=0$, $x_1=1$ and $x_n=2x_{n-1}+x_{n-2}$ for all $n\geq 2$. Then $x_{100}=$ _____.

(5) Consider the following system of linear equations: $\begin{cases}x_1+2x_2-2x_3=0\\2x_1-x_2+ax_3=0\\3x_1+x_2-x_3=0\end{cases}$

If the columns of a nonzero 3x3 matrix B are sdutions to the above system, then $a=$ _____, $|B|=$ _____.

3. Consider the 5x5 matrix $A=\begin{bmatrix}3&5&8&1&-6\\-3&-4&-7&1&9\\3&4&7&-1&-9\\-3&-4&-7&1&9\\2&3&5&0&5\end{bmatrix}$

The characteristic polynomial of A is $p(x)=-x^3(x-1)^2$. And the reduced row echelon from of $A-I$ is given by $A-I\to\begin{bmatrix}1&0&0&-1&-3\\0&1&0&-1&0\\0&0&1&1&0\\0&0&0&0&0\\0&0&0&0&0\end{bmatrix}$

(a) Find an invertible matrix S such that $S^{-1}AS$ is diagonal.

(b) Compute A^k for any positive integer k.

4. Find the matrix Q in the QR-factorization of the matrix $A=\begin{bmatrix}1&1&0\\1&0&1\\0&1&1\end{bmatrix}$.

5. Let $A=\begin{bmatrix}0&-i&0\\i&1&i\\0&-i&0\end{bmatrix}$

(a) Show that A is Hermitian.

(b) Find the eigenvalues and eigenvectors of A.

(c) Can we find a unitary matrix U such that U^HAU is diagonal? If yes, find one

such matrix. Otherwise, explain.

6. Let B be an $n\times n$ real symmetric matrix. A quadratic form $g(y)=y^TBy$ is called negative ~~define~~ definite if $y^TBy<0$ for all nonzero real vectors $y\in R^n$. Consider the following quadratic form

$$f(x_1,x_2,x_3)=(\lambda-3)x_1^2+4x_1x_2+\lambda x_2^2+(\lambda-1)x_3^2$$

(a) Find a real symmetric matrix A, such that $f(x_1,x_2,x_3)=x^TAx$, where $x=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$.

(b) Determine the range of λ such that $f(x_1,x_2,x_3)$ is negative definite.

7. Let A,B be two $n\times n$ real matrices. Suppose A has n distinct eigenvalues and $AB=BA$.

(a) Show that B is diagonalizable.

(b) Suppose $n=3$. Show that there exists a polynomial of degree at most 2, $g(x)$, such that $B=f(A)$.

(c) Suppose $n>3$. Show that there exists a polynomial of degree at most $n-1$, $f(x)$, such that $B=f(A)$.